

Q1. First Law of Thermodynamics.

The **First Law of Thermodynamics** states that:

"Energy can neither be created nor destroyed. It can only be transformed from one form to another."

When heat is supplied to a system, it is used to:

- Increase the **internal energy** of the system.
- Perform **work** on the surroundings.

Mathematical Expression

If

- Q = Heat supplied to the system
- W = Work done by the system
- ΔU = Change in internal energy

Then,

$$Q = \Delta U + W$$

or

$$\boxed{\Delta U = Q - W}$$

This is the **mathematical form of the First Law of Thermodynamics**.

Applications of the First Law

1. Isochoric (Constant Volume) Process

Since volume remains constant,

$$W = 0$$

Therefore,

$$Q = \Delta U$$

Conclusion: All supplied heat increases internal energy.

2. Isothermal (Constant Temperature) Process

Temperature remains constant.

Hence,

$$\Delta U = 0$$

Therefore,

$$Q = W$$

Conclusion: All supplied heat is converted into work.

3. Adiabatic Process

No heat enters or leaves the system.

$$Q = 0$$

Therefore,

$$\Delta U = -W$$

Conclusion: Work is done at the expense of internal energy.

4. Cyclic Process

After one complete cycle,

$$\Delta U = 0$$

Therefore,

$$Q = W$$

Conclusion: Net heat supplied equals net work done.

Diagram

Limitations

1. It does **not indicate the direction** of heat flow.
2. It does **not explain why heat flows from hot to cold**.
3. It **cannot determine whether a process is reversible or irreversible**.
4. It **does not provide the maximum possible efficiency** of heat engines.

Importance

- Forms the basis of thermodynamics.
- Used in studying heat engines, refrigerators, and air conditioners.
- Explains the principle of conservation of energy in thermal processes.
- Widely applied in physics and engineering.

Q2. Second Law of Thermodynamics.

The **Second Law of Thermodynamics** states that:

"Heat cannot flow spontaneously from a colder body to a hotter body without external work."

It explains the **direction of heat flow** and shows that not all heat can be converted into useful work.

Kelvin–Planck Statement

According to Kelvin–Planck:

"It is impossible to construct a heat engine that converts all the heat absorbed from a source completely into work without rejecting some heat to a sink."

Explanation

- A heat engine absorbs heat Q_1 from a hot reservoir.
- It converts only part of this heat into work W .
- The remaining heat Q_2 is rejected to a cold reservoir.

Therefore,

$$Q_1 = W + Q_2$$

Since $Q_2 \neq 0$, **100% efficiency is impossible**.

Clausius Statement

According to Clausius:

"Heat cannot, by itself, flow from a colder body to a hotter body."

Explanation

- Heat naturally flows from **hot → cold**.
- To transfer heat from **cold → hot**, external work is required.

Example: Refrigerator and Air Conditioner.

Mathematical Representation

For a heat engine,

$$Q_1 = W + Q_2$$

where:

- Q_1 = Heat absorbed from the hot reservoir
- Q_2 = Heat rejected to the cold reservoir
- W = Work done by the engine

Efficiency of the engine:

$$\eta = \frac{W}{Q_1}$$

Since $Q_2 > 0$,

$$\eta < 1$$

Thus, **no heat engine can have 100% efficiency**.

Diagram

Importance of the Second Law

1. Explains the **natural direction of heat flow**.
2. Shows that **complete conversion of heat into work is impossible**.
3. Forms the basis for the design of **heat engines**.
4. Explains the working of **refrigerators and air conditioners**.
5. Introduces the concept of **entropy**, which measures the disorder or randomness of a system.

Applications

1. Heat Engines

Steam engine, petrol engine , diesel engine

Only a part of heat is converted into work.

2. Refrigerators

Heat is transferred from a cold region to a hot region by supplying external work.

3. Air Conditioners

Operate on the same principle as refrigerators by removing heat from indoor air and releasing it outside.

Limitations

1. It does **not describe the amount of heat transferred**.
2. It does **not provide the exact efficiency** of all engines.
3. It **cannot replace the First Law**, which deals with conservation of energy.

Q3. Define Entropy in thermodynamics.

Entropy is a thermodynamic property that measures the **degree of disorder, randomness, or unavailability of energy** in a system.

It is represented by the symbol S .

Simple Definition (for exams):

Entropy is the measure of randomness or disorder of a thermodynamic system.

Mathematical Expression

For a reversible process,

$$\boxed{dS = \frac{\delta Q_{rev}}{T}}$$

or, for a finite change,

$$\boxed{\Delta S = \frac{Q_{rev}}{T}}$$

where,

- S = Entropy
- Q_{rev} = Heat absorbed reversibly
- T = Absolute temperature (Kelvin)

SI Unit

$$\boxed{\text{Joule per Kelvin (J/K)}}$$

Change in Entropy

1. Reversible Process

$$\Delta S = \frac{Q_{rev}}{T}$$

Entropy changes gradually.

2. Irreversible Process

Entropy of the universe **increases**.

3. Adiabatic Reversible Process

Since,

$$Q = 0$$

Therefore,

$$\Delta S = 0$$

Entropy remains constant (an **isentropic process**).

Properties of Entropy

1. Entropy is a **state function**.
2. It depends only on the **initial and final states** of the system.
3. Its SI unit is **J/K**.
4. Entropy increases with increasing temperature.
5. Entropy increases when a solid melts or a liquid evaporates.
6. In an isolated system, entropy never decreases.

Importance of Entropy

1. Determines the direction of spontaneous processes.
2. Explains why heat flows naturally from hot to cold.
3. Helps calculate the efficiency of heat engines.
4. Indicates the degree of disorder in a system.
5. Plays a key role in thermodynamics and engineering.

Applications

Heat engines, Refrigerators, Air conditioners, Steam power plants, Chemical reactions, Industrial thermodynamic processes

Advantages: **1.** Measures randomness in a system. **2** Predicts whether a process is spontaneous. **3.**Helps determine the maximum efficiency of engines. **4.**Useful in engineering and physical chemistry.

Limitations

1. Entropy is an abstract quantity and cannot be measured directly.
2. It alone cannot determine all thermodynamic properties.
3. It does not indicate the speed of a process.

Q3. Carnot Engine.

A **Carnot engine** is an **ideal reversible heat engine** that operates between two heat reservoirs at different temperatures. It converts part of the heat absorbed from the hot reservoir into work and rejects the remaining heat to the cold reservoir.

Principle

A Carnot engine works on the **Carnot cycle**, which consists of **four reversible thermodynamic processes**:

1. Isothermal Expansion , 2. Adiabatic Expansion 3. Isothermal Compression . 4. Adiabatic Compression

Construction

A Carnot engine consists of: 1. A cylinder containing an ideal gas. 2. A frictionless movable piston. 3. A hot reservoir at temperature T_1 . 4. A cold reservoir at temperature T_2 .

Working of Carnot Cycle

1. Isothermal Expansion (A → B) 1. The gas is kept in contact with the **hot reservoir**. 2. It absorbs heat Q_1 . 3. Temperature remains constant at T_1 . 4. The gas expands and performs work.

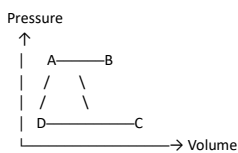
2. Adiabatic Expansion (B → C): 1. The cylinder is insulated. 2. No heat enters or leaves the system. $Q = 0$. 3. The gas continues to expand. 4. Its temperature decreases from T_1 to T_2 .

3. Isothermal Compression (C → D): 1. The gas is placed in contact with the **cold reservoir**. 2. Temperature remains constant at T_2 . 3. Heat Q_2 is rejected to the cold reservoir.

4. Adiabatic Compression (D → A): 1. The cylinder is insulated again. 2. No heat exchange takes place. 3. External work is done on the gas. 4. Temperature rises back from T_2 to T_1 . 5. The engine returns to its initial state, completing one cycle.

Diagram

Carnot Cycle (P–V Diagram)



AB → Isothermal Expansion , BC → Adiabatic Expansion , CD → Isothermal Compression , DA → Adiabatic Compression

Efficiency of Carnot Engine

Efficiency is defined as: $\eta = \frac{\text{Work Done}}{\text{Heat Supplied}}$

Since, $W = Q_1 - Q_2$

Therefore, $\eta = \frac{Q_1 - Q_2}{Q_1}$

or, $\eta = 1 - \frac{Q_2}{Q_1}$

For a reversible Carnot engine,

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2}$$

Hence, $\eta = 1 - \frac{T_2}{T_1}$

where, T_1 = Temperature of hot reservoir (Kelvin) , T_2 = Temperature of cold reservoir (Kelvin)

Applications: Steam power plants , Internal combustion engines , Refrigerators , Air conditioners , Thermal power stations

Advantages: 1. Maximum possible efficiency. 2. Completely reversible cycle. 3. No energy is wasted due to friction or irreversibility. 3. Acts as an ideal model for all heat engines.

Limitations: 1. It is an **ideal engine** and cannot be built perfectly in practice. 2. All processes must be **reversible**, which is not possible in real life. 3. Heat transfer at an infinitesimally small temperature difference requires a very long time.

Q4. Black Body Radiation.

A **black body** is an ideal body that **absorbs all the incident radiation** falling on it, regardless of its wavelength or direction, and **emits the maximum possible radiation** at a given temperature.

The radiation emitted by a black body is called **black body radiation**.

Characteristics of a Black Body

1. Absorbs **100%** of the incident radiation. 2. Reflects **no** radiation. 3. Transmits **no** radiation. 4. Emits the maximum possible thermal radiation. 5. The emitted radiation depends **only on its temperature**, not on the material.

Construction of a Black Body

A practical black body is made using:

1. A hollow metal cavity. 2. A very small hole on one side. 3. The inner surface is coated with black soot.

When radiation enters through the small hole, it undergoes multiple reflections inside the cavity and is almost completely absorbed. The small hole therefore behaves like a perfect black body.

Diagram:

Black Body Radiation

When a black body is heated, it emits electromagnetic radiation called **black body radiation**.

Properties

1. It depends only on the **temperature** of the body. 2. As temperature increases: A) The amount of emitted radiation increases. B) The wavelength corresponding to maximum intensity decreases. 3. The radiation consists of a continuous spectrum.

Black Body Radiation Spectrum (graph)

Observation:

1. Higher temperature → Higher peak intensity. 2. Higher temperature → Peak shifts toward shorter wavelengths.

Important Laws of Black Body Radiation

1. Stefan–Boltzmann Law

The total energy radiated per second is proportional to the fourth power of absolute temperature.

$$E = \sigma T^4$$

where, E = Energy emitted per unit area , σ = Stefan–Boltzmann constant , T = Absolute temperature (Kelvin)

2. Wien's Displacement Law

The wavelength at which maximum radiation occurs is inversely proportional to the absolute temperature.

$\lambda_{max} T = b$ where, λ_{max} = Wavelength of maximum intensity , T = Absolute temperature , b = Wien's constant

3. Planck's Radiation Law

Planck explained black body radiation by proposing that energy is emitted in small packets called **quanta**.

The energy of each quantum is:

$$E = h\nu \text{ where, } E = \text{Energy, } h = \text{Planck's constant } \nu = \text{Frequency}$$

This theory successfully explained the complete black body radiation spectrum.

Applications: 1. Measuring the temperature of stars. 2. Designing furnaces and industrial heaters. 3. Infrared cameras and thermal imaging. 3. Radiation thermometers (Pyrometers). 4. Space science and astrophysics.

Advantages: 1. Serves as the ideal emitter and absorber of radiation. 2. Helps explain thermal radiation. 3. Forms the basis of quantum theory. 4. Widely used in astronomy and engineering.

Limitations: 1. A perfect black body does not exist in nature. 2. It is an ideal theoretical model. 3. Practical black bodies only approximate its behavior.

Q5. Define Maxwell-Boltzmann Distribution.

The **Maxwell–Boltzmann Distribution Law** describes the **distribution of the speeds (or energies) of gas molecules** in an ideal gas at a given temperature.

It states that: "In an ideal gas, molecules move with different speeds. At a given temperature, only a few molecules have very low or very high speeds, while most molecules have an intermediate (most probable) speed."

Maxwell–Boltzmann Distribution Law

The mathematical form is:

$$f(v) = 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} v^2 e^{-mv^2/2kT}$$

where, $f(v)$ = Fraction of molecules having speed v , m = Mass of one molecule , k = Boltzmann constant , T = Absolute temperature (Kelvin) , v = Speed of the molecule

Distribution Curve (graph):

Observation: 1. At **low temperature (T_1)**: a) Curve is taller and narrower.

2. At **high temperature (T_2)**: a) Curve becomes broader and shifts to the right. 3. As temperature increases, the **most probable speed increases**.

Types of Molecular Speeds

1. Most Probable Speed (v_p)

The speed possessed by the **largest number of molecules**.

$$v_p = \sqrt{\frac{2kT}{m}}$$

2. Average Speed (v_{avg})

The arithmetic mean of all molecular speeds.

$$v_{avg} = \sqrt{\frac{8kT}{\pi m}}$$

3. Root Mean Square (RMS) Speed (v_{rms})

The square root of the average of the squares of molecular speeds.

$$v_{rms} = \sqrt{\frac{3kT}{m}}$$

Relation Between Speeds

$$v_p < v_{avg} < v_{rms}$$

This relation is **very important** and is often asked as a **1-mark or 2-mark question**.

Applications 1. Explains the motion of gas molecules. 2. Used to calculate molecular speeds. 3. Helps explain diffusion and effusion of gases. 4. Used in kinetic theory of gases. 5. Important in thermodynamics and statistical mechanics.

Advantages: 1. Explains the distribution of molecular speeds. 2. Supports the kinetic theory of gases. 3. Predicts average, RMS, and most probable speeds. 4. Helps understand gas behavior at different temperatures.

Limitations: 1. Applicable only to ideal gases. 2. Fails at very low temperatures. 3. Does not explain the behavior of bosons or fermions. 4. Not valid for quantum particles.

Conclusion

The Maxwell–Boltzmann Distribution explains how the speeds of gas molecules are distributed in an ideal gas. It shows that molecules have different speeds and that most molecules move with the most probable speed. As temperature increases, molecular speeds also increase. The distribution is fundamental to the kinetic theory of gases and statistical mechanics.

Q6. Define Maxwell Relations.

Maxwell Relations are a set of four thermodynamic equations derived from the thermodynamic potentials. They establish relationships between measurable thermodynamic quantities such as pressure (P), volume (V), temperature (T), and entropy (S).

Thermodynamic Potentials

The four thermodynamic potentials are: Internal Energy (U), Enthalpy (H), Helmholtz Free Energy (F or A), Gibbs Free Energy (G)

These potentials are used to derive the four Maxwell relations.

1. First Maxwell Relation

From Internal Energy (U), $dU = TdS - PdV$

Therefore, $\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V$

2. Second Maxwell Relation

From Enthalpy (H), $dH = TdS + VdP$

Therefore, $\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$

3. Third Maxwell Relation

From Helmholtz Free Energy (F or A), $dF = -SdT - PdV$

Therefore, $\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$

4. Fourth Maxwell Relation

From Gibbs Free Energy (G), $dG = -SdT + VdP$

Therefore, $\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P$

Diagram (Thermodynamic Potentials)

Thermodynamic Potentials	
Internal Energy (U)	
$dU = TdS - PdV$	
First Maxwell Relation	
Enthalpy (H)	Helmholtz Free Energy (F)
$dH = TdS + VdP$	$dF = -SdT - PdV$
Second Maxwell Relation	Third Maxwell Relation
Gibbs Free Energy (G)	
$dG = -SdT + VdP$	
Fourth Maxwell Relation	

Applications: 1. Calculation of entropy changes. 2. Study of heat engines and refrigerators. 3. Chemical thermodynamics. 4. Statistical mechanics. 4. Engineering and power plant analysis.

Advantages: 1. Reduce complex calculations. 2. Relate measurable properties such as pressure, volume, and temperature. 3. Useful in solving thermodynamic problems. 4. Form the foundation of advanced thermodynamics.

Limitations 1. Applicable only to systems in thermodynamic equilibrium. 2. Not directly applicable to irreversible processes. 3. Require thermodynamic potentials for derivation.

Conclusion

The **Maxwell Relations** are four important equations obtained from the thermodynamic potentials. They connect pressure, volume, temperature, and entropy through partial derivatives, making many thermodynamic calculations easier. These relations are fundamental in the study of thermodynamics and have wide applications in physics, chemistry, and engineering.

Q7. Define Thermodynamic Potentials.

Thermodynamic Potentials are thermodynamic quantities that describe the **energy state of a system** under different conditions. They help determine the direction of thermodynamic processes and the equilibrium state of a system.

They are obtained from the **First and Second Laws of Thermodynamics**.

There are **four thermodynamic potentials**: 1. Internal Energy (U) 2. Enthalpy (H) 3. Helmholtz Free Energy (F or A) 4. Gibbs Free Energy (G)

1. Internal Energy (U) : internal energy is the **total energy** possessed by a system due to the random motion and interactions of its molecules.

Equation $dU = TdS - PdV$

where, T= Temperature, S= Entropy, P= Pressure, V= Volume

Importance: 1. Represents the total energy stored in a system. 2. Used in the **First Law of Thermodynamics**.

2. Enthalpy (H): Enthalpy is the **total heat content** of a system.

It is defined as: $H = U + PV$

Differential Form: $dH = TdS + VdP$

Importance: 1. Useful in constant-pressure processes. 2. Used to calculate heat changes in chemical reactions.

3. Helmholtz Free Energy (F or A): Helmholtz free energy is the amount of energy available to perform **useful work** at constant temperature and volume.

Equation $F = U - TS$

Differential Form: $dF = -SdT - PdV$

Importance: 1. Used in systems maintained at constant temperature and volume. 2. Helps determine spontaneous processes under these conditions.

4. Gibbs Free Energy (G): Gibbs free energy is the energy available to perform useful work at **constant temperature and pressure**.

Equation $G = H - TS$ Or, $G = U + PV - TS$

Differential Form: $dG = -SdT + VdP$

Importance: 1. Predicts whether a process is spontaneous. 2. Widely used in chemistry and engineering.

Applications: Heat engines, Refrigerators, Chemical reactions, Power plants, Engineering thermodynamics, Statistical mechanics

Advantages: 1. Simplify energy calculations. 2. Predict spontaneous processes. 3. Describe systems under different conditions. 4. Help derive important thermodynamic equations.

Limitations: 1. Applicable mainly to systems in thermodynamic equilibrium. 2. Difficult to measure directly. 3. Some potentials are useful only under specific conditions.

Diagram:

Q8. Define Equipartition of Energy

The **Law (or Theorem) of Equipartition of Energy** states that: "At thermal equilibrium, the total energy of a system is equally distributed among all its independent degrees of freedom, and each degree of freedom possesses an average energy of $\frac{1}{2} kT$."

Degrees of Freedom

Degrees of Freedom (f) are the **minimum number of independent ways in which a molecule can move or store energy**.

They include: **Translational motion** (movement along x, y, z axes), **Rotational motion**, **Vibrational motion**

Average Energy per Degree of Freedom

Each degree of freedom contributes: $\frac{1}{2} kT$

where, k= Boltzmann constant, T= Absolute temperature (Kelvin)

Total Energy of One Molecule: If a molecule has f degrees of freedom, then $E = \frac{f}{2} kT$

where, E= Average energy of one molecule f= Degrees of freedom

Total Energy of One Mole: For one mole of gas, $E = \frac{f}{2} RT$

where, R= Universal gas constant T= Absolute temperature

Degrees of Freedom of Different Molecules

Molecule	Degrees of Freedom (f)	Energy
Monoatomic Gas (He, Ne, Ar)	3	$\frac{3}{2} kT$
Diatomic Gas (O ₂ , N ₂)	5	$\frac{5}{2} kT$
Non-linear Polyatomic Gas	6	$3kT$

Note: At ordinary temperatures: 1. Monoatomic gases → **3 degrees of freedom**, 2. Diatomic gases → **5 degrees of freedom**

Monoatomic Gas: A monoatomic molecule moves only in three directions: x-axis, y-axis **and** z-axis

Therefore, f = 3

Average energy: $E = \frac{3}{2} kT$

Diatomic Gas: A diatomic molecule possesses: 1. 3 translational degrees 2. 2 rotational degrees

Therefore, f = 5

Average energy: $E = \frac{5}{2} kT$

Importance: 1. Explains the distribution of molecular energy. 2. Forms the basis of the kinetic theory of gases. 3. Helps calculate internal energy. 4. Explains the heat capacities of gases. 5. Useful in thermodynamics and statistical mechanics.

Applications: Ideal gas calculations, Internal energy of gases, Specific heat calculations, Statistical mechanics, Molecular physics

Advantages: 1. Simple method for calculating molecular energy. 2. Explains heat capacities of gases. 3. Useful in kinetic theory. 4. Applicable to ideal gases at ordinary temperatures.

Limitations: 1. Valid mainly for **ideal gases**. 2. Fails at **very low temperatures**, where quantum effects become important. 3. Does not correctly explain the heat capacities of all real gases. 4. Ignores quantum mechanical behavior.

Conclusion

The **Equipartition of Energy Theorem** states that the total energy of a system is equally shared among all its degrees of freedom, with each degree contributing an average energy of $\frac{1}{2} kT$. It is a fundamental result of the **kinetic theory of gases** and is widely used in thermodynamics to calculate the internal energy and heat capacities of gases.

Q9. Plancks Radiation Law

"A hot body emits or absorbs radiant energy discontinuously in the form of small packets of energy called quanta (or photons)."

The energy of each quantum is directly proportional to the frequency of radiation.

Planck's Quantum Theory (Assumptions)

Planck proposed the following assumptions: 1.Energy is emitted or absorbed in **discrete packets** called **quanta**. 2.Each quantum has a definite amount of energy. 3.The energy of a quantum is proportional to its frequency. 4.Molecules emit or absorb energy only in whole-number multiples of a quantum. 5.Energy is **not emitted continuously**.

Planck's Equation

The energy of one quantum is: $E = h\nu$

where, E = Energy of one photon (Joule) h = Planck's constant $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$, ν = Frequency of radiation (Hz)

Meaning of the Equation

From, $E = h\nu$

we conclude: 1. Higher frequency $\rightarrow$ Higher energy 2.Lower frequency $\rightarrow$ Lower energy

Thus, **energy is directly proportional to frequency**.

Planck's Radiation Formula: The spectral energy density of black body radiation is given by: $u(\nu) = \frac{8\pi h\nu^3}{c^3} \cdot \frac{1}{e^{h\nu/kT} - 1}$

where, $u(\nu)$ = Energy density, h = Planck's constant, c = Speed of light, k = Boltzmann constant, T = Absolute temperature

Black Body Radiation Curve:

Higher Temperature $\rightarrow$ Higher Intensity

Higher Temperature $\rightarrow$ Shorter Peak Wavelength

Applications: Black body radiation analysis. Quantum mechanics. Laser technology. Semiconductor devices. Astronomy and astrophysics. Thermal imaging. Spectroscopy.

Advantages: 1.Correctly explains black body radiation. 2.Introduces the photon concept. 3.Agrees with experimental observations. 4.Foundation of quantum physics.

Limitations: 1.It is mainly applicable to black body radiation. 2.It does not describe all quantum phenomena by itself. 3.More advanced quantum mechanics is needed for microscopic systems.

Conclusion:

Planck's Radiation Law states that energy is emitted or absorbed in the form of **discrete quanta**, each having energy: $E = h\nu$

This theory successfully explained **black body radiation** and marked the beginning of **quantum physics**. It remains one of the most important concepts in modern physics.

Q10. Isothermal and Adiabatic Processes (derivations)

1. Isothermal Process (Derivation): An isothermal process is a thermodynamic process in which the **temperature remains constant** throughout the process. $T = \text{Constant}$

For an ideal gas, $PV = nRT$

Derivation of Work Done

The work done during expansion is, $W = \int_{V_1}^{V_2} P \, dV$

Hence, $W = nRT \ln \left(\frac{V_2}{V_1} \right)$

Internal Energy, Since temperature remains constant, $\Delta U = 0$

From the First Law, $Q = \Delta U + W$

Therefore, $Q = W$

Conclusion: All the heat supplied is converted into work.

P–V Diagram:

The curve is called an **Isothermal Curve**.

Characteristics: 1.Temperature remains constant. 2. $PV = \text{Constant}$. 3.Internal energy does not change. 4.Heat supplied equals work done

2. Adiabatic Process (Derivation): An **adiabatic process** is a thermodynamic process in which **no heat enters or leaves the system**. $Q = 0$

The equation for an adiabatic process is, $PV^\gamma = \text{Constant}$

Where, $\gamma = \frac{C_p}{C_v}$ is called the **adiabatic index**.

Derivation of Work Done

Since, $PV^\gamma = C$

where C is a constant, $P = \frac{C}{V^\gamma}$

Work done is, $W = \int_{V_1}^{V_2} P \, dV$

Substitute P , $W = \int_{V_1}^{V_2} \frac{C}{V^\gamma} \, dV$

Integrating, $W = \frac{C}{1-\gamma} (V_2^{1-\gamma} - V_1^{1-\gamma})$

Since, $C = PV^\gamma$

the expression becomes: $W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$

This is the required expression for work done in an adiabatic process.

Internal Energy, Since $Q = 0$

From the First Law, $Q = \Delta U + W$

Therefore, $0 = \Delta U + W$

Or, $\Delta U = -W$

Conclusion: Work is done at the expense of internal energy.

P–V Diagram:

The adiabatic curve is **steeper** than the isothermal curve.

Applications

Isothermal Process: 1.Slow expansion of gases. 2.Industrial gas compression. 3.Laboratory experiments.

Adiabatic Process: 1.Diesel and petrol engines. 2.Air compressors. 3.Refrigerators. 4.Atmospheric expansion.

Conclusion:

Isothermal Process: Temperature remains constant, and all heat supplied is converted into work. $W = nRT \ln \left(\frac{V_2}{V_1} \right)$

Adiabatic Process: No heat is exchanged with the surroundings, and work is done by changing the internal energy. $W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$

Q11. Wien's Displacement Law

Wien's Displacement Law states that: "The wavelength corresponding to the maximum intensity of radiation emitted by a black body is inversely proportional to its absolute temperature." , **Mathematical Statement:** $\lambda_{\text{max}} \propto \frac{1}{T}$ or, $\lambda_{\text{max}} T = b$

where, λ_{max} = Wavelength of maximum intensity (m) , T = Absolute temperature (K) , b = Wien's displacement constant

Wien's Constant: $b = 2.898 \times 10^{-3} \text{ m}\cdot\text{K}$

Graph:

Higher Temperature $\rightarrow$ Shorter Peak Wavelength

Significance: 1.Shows the relationship between temperature and peak wavelength. 2.Explains why the color of a hot object changes with temperature. 3.Helps determine the surface temperature of stars. 4.Supports the study of black body radiation.

Applications: 1.Measuring the temperature of stars and the Sun. 2.Infrared thermometers. 3.Thermal imaging cameras. 4.Astronomy and astrophysics. 5.Industrial temperature measurement.

Advantages: 1.Simple relationship between wavelength and temperature. 2. Useful for determining very high temperatures. 3.Importance in astronomy and thermal physics.

Limitations: 1.Applicable only to **black bodies** or objects that closely behave like black bodies. 2.Does not describe the complete radiation spectrum. 3.Gives only the wavelength of **maximum intensity**, not the total emitted energy.

Example:The surface temperature of the Sun is about **5800 K**.

Using Wien's Law: $\lambda_{\text{max}} = \frac{2.898 \times 10^{-3}}{5800} \approx 5 \times 10^{-7} \text{ m}$

This lies in the **visible region**, which is why sunlight appears visible to the human eye.

Conclusion : Wien's Displacement Law states that the wavelength corresponding to the maximum intensity of black body radiation is **inversely proportional** to its absolute temperature.

The mathematical expression is: $\lambda_{\text{max}} T = b$

This law is widely used in **astronomy**, **thermal imaging**, and **temperature measurement**.

Q12. Gibbs Free Energy

Gibbs Free Energy is the amount of energy available in a system to perform **useful work** at **constant temperature and pressure**.

It is represented by the symbol G .

The Gibbs Free Energy is given by: $G = H - TS$

where, G = Gibbs Free Energy, H = Enthalpy, T = Absolute temperature (Kelvin) , S = Entropy

Another form is: $G = U + PV - TS$

where, U = Internal Energy , P = Pressure , V = Volume

Differential Form: $dG = -SdT + VdP$

Conditions for Spontaneity: 1. $\Delta G < 0 \rightarrow$ Process is spontaneous. 2. $\Delta G = 0 \rightarrow$ System is at **equilibrium**. 3. $\Delta G > 0 \rightarrow$ Process is **non-spontaneous**.

Applications: 1.Predicts spontaneous chemical reactions. 2.Used in thermodynamics and chemistry. 3.Helps determine equilibrium conditions. 4.Used in engineering and industrial processes.

Advantages: 1.Predicts the feasibility of a process. 2.Useful for constant temperature and pressure conditions. 3.Widely applied in chemistry and thermodynamics.

Limitations: 1.Applicable mainly at **constant temperature and pressure**. 2.Does not indicate the **speed** of a reaction. 3.It predicts only whether a process is possible, not how fast it occurs.

Conclusion: Gibbs Free Energy is the energy available to perform useful work at constant temperature and pressure. It is an important thermodynamic potential used to predict the **spontaneity** and **equilibrium** of physical and chemical processes.